Parth Kishorbhai Vekariya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.15
SHREE DHARMAJIVAN SWAMI SECONDARY AND HIGHER SECONDARY SCHOOL	Class XII (GSEB)	2021	90.30%
SHREE SWAMINARAYAN GURUKUL VIDHYALAYA TARAVADA	Class X (GSEB)	2019	81.16%

WORK EXPERIENCE

Shreeji Software | Software Intern

- **Gathered client requirements** and coordinated implementation for a solar investment platform serving **100+** customers.
- Designed **MySQL** schemas and **revenue workflows** for investments, ownership shares, monthly earnings, and automated investor reports.

CSRBOX - IBM SkillsBuild | Data Analytics Intern

- **Analyzed** and **cleaned** AtliQ sales data using **Excel**, preparing transactional data for structured business analysis.
- Built interactive **Power BI dashboards** with **data modeling** and **calculated measures**, **analyzing** revenue, sales quantity, profit margins, markets, customers, and products.
- Generated **sales** and **profitability insights** through market, customer, product, and time-based analysis, tracking **Rs. 336M** revenue, **847K** sales quantity, and **Rs. 10.5M** total profit margin.

PROJECTS

Distributed GPU Resource Sharing Platform

- Built a **P2P GPU-sharing platform** using **WebSockets** and **Python** to monetize idle compute and provide **affordable GPU access**.
- Developed real-time **resource monitoring** for **GPU, VRAM, CPU, RAM, storage**, and **network** with heartbeat-based node tracking.
- Secured workloads using **Docker isolation** and strict command allowlisting, preventing **host-level access**.

Llama 3.2 3B LLM Fine-Tuning with QLoRA

- Fine-tuned a **3B Llama-3.2-Instruct** LLM using **QLoRA, 4-bit quantization,** and **Unsloth** for math and logic reasoning.
- Trained on **171K+** reasoning examples with custom <think> formatting, **Transformers,** and **TRL SFTTrainer**.
- Applied **rank-16 LoRA** adapters across attention/MLP layers with **8-bit AdamW**, gradient accumulation, and mixed precision.
- Exported to **Q8_0 GGUF** and integrated with **Ollama** for local LLM inference.

Crop Disease Advisor

- Built an AI system for crop disease detection and treatment planning using **EfficientNet-B4 + Qwen2.5-3B QLoRA** across **38 crop disease classes**.
- Trained EfficientNet-B4 with 3-phase unfreezing, FP16 AMP, and CosineAnnealingWarmRestarts, achieving **96.8% accuracy, 95.4% macro F1**, and **97.1% AUC**.
- Fine-tuned **Qwen2.5-3B** using **4-bit NF4 QLoRA + TRL SFTTrainer**, achieving **81.68% BERTScore F1**.

Student Performance Analytics & Prediction

- Built a prediction pipeline for math-score prediction across **1,000 records**.

- Preprocessed **7 features** using **imputation, OneHotEncoder**, and **StandardScaler**.
- Benchmarked **7 regression** models with **GridSearchCV** and **3-fold cross-validation**, including **XGBoost, CatBoost**, and **Random Forest**.
- Achieved **0.8804 R²** on **200** holdout records, with **Linear Regression** performing best.

AI Operations Command Center

- Built a multi-agent AI system using **LangGraph** for stateful workflow orchestration.
- Engineered an **MCP tool** layer with **14 tools** for secure LLM interaction with **11+** MongoDB collections.
- Developed a **RAG** pipeline using **chunking, embeddings,** and **semantic retrieval** for knowledge-based Q&A.

POSITIONS OF RESPONSIBILITY

MSME Tech Connect 2026 | Head of Visitor & Exhibitor Relations

- **Led outreach to IITs, NITs,** and technology cells to onboard exhibitors.
- **Managed 100+ exhibitors**, handling selection, stall allocation, and event coordination.
- **Led 10+** coordinators and volunteers, **managing visitor registration**, footfall tracking, and stall support.

RELEVANT COURSES

Technical: Data Science, Big Data Analytics, Data Analysis and Visualization, Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Network Security, Machine Learning

Management: Strategy Management, Operations Management, Financial Planning & Budgeting, Marketing, Design Thinking, HR & Leadership

SKILLS

Programming: Python, SQL, C++ | **ML:** Machine Learning, Statistics, Scikit-learn, PyTorch | **GenAI:** LLMs, Transformers, QLoRA, LoRA, Hugging Face, RAG, LangGraph, MCP | **Data:** Pandas, NumPy, Power BI, Excel | **Backend/DB:** FastAPI, PostgreSQL | **DevOps & Tools:** Docker, GitHub Actions, Kubernetes, Git, Ollama

EXTRACURRICULAR

Kyoto University Student Immersion Program: Worked with **4 Kyoto University** students for **7 days** to validate a Japan-India energy problem and develop an AI-based solution using weather and peak-hour electricity forecasting to optimize building cooling schedules and **reduce energy costs**.

Tongali Entrepreneurship Program: Developed an India-focused **GTM strategy and business model** for a Japanese EdTech startup, collaborating directly with a Japanese entrepreneur during a **3-day venture immersion** and presenting the strategy to industry experts and mentors.